

Applying Prompt-based Mitigation of Gender-Role Bias in Large Language Models for Security

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Abstract—As large language models (LLMs) become increasingly integrated into society, concerns over fairness and bias have emerged as critical areas of research. Prior research has demonstrated that LLMs often reflect stereotypes present in their training data, and while this is informative, systematic studies producing rigorous experiments and tangible measurements remain limited. Furthermore, as LLMs are deployed in security-sensitive domains, such as industrial control systems (ICS), Internet-of-Things (IoT) networks, and critical infrastructure, exploring tangible, quantitative bias mitigation techniques becomes essential for real-time deployment. In this work, as a proof of concept, we apply our pipelines to 53 gender-neutral professional resume prompts to generally analyze how LLMs assign gender-specific attributes. Using these prompts, we implement four distinct mitigation pipelines: a Baseline with no intervention, In-Context Example prompting, Self-Evaluation and Regeneration, and Chain-of-Thought (CoT) reasoning. Furthermore, to quantify each pipeline, we introduce the Stereotype Alignment Rate (SAR) to quantify bias, defined as the percentage of outputs conforming to traditional gender stereotypes. Results across each experimental pipeline indicate an effective reduction in bias, lowering the SAR by up to 15% compared to the baseline. This work demonstrates that prompt engineering, particularly through the use of counter-stereotypical examples, can serve as an effective and lightweight method for mitigating bias in LLM outputs.

Index Terms—Large Language Models (LLMs), GPT-4, bias mitigation, ethical AI.

I. INTRODUCTION

The artificial intelligence revolution, currently led by transformer-based frameworks [1] and large language models (LLMs) [2], have become central to a wide range of modern applications across societal domains, spanning from autonomous vehicle applications [3], [4] to job vacancy filtering and recommendation systems [5], [6], and even to security domains such as critical infrastructure protection [7], [8]. By training on massive web-scraped datasets, these models learn the contextual relationships between words and predict the next token in a sequence, resulting in remarkable fluency [9]. However, because pretraining data is drawn from uncured, web-scale sources, the resulting training distributions not only capture meaningful linguistic patterns but also encode latent, real-world biases [10]. Biases embedded in these model's weights can be harmful, including those based on gender, race, and socioeconomic status. Moreover, bias in security-critical outputs, such as applications in cybersecurity, could

leave networks under-protected against a wide range of attacks. For instance, when automating the creation of firewall rules or intrusion-detection signatures for industrial control system (ICS) and Internet of Things (IoT) deployments, an unmitigated model may overemphasize common attack patterns seen during training, leaving rare or novel exploits insufficiently covered. As LLMs are deployed across sensitive domains, the presence of bias in their outputs raise critical concerns regarding fairness, accountability, and trustworthiness.

A striking early example of LLM bias occurred with Microsoft's Tay chatbot, which rapidly adopted and amplified offensive language after interacting with users on Twitter [11]. This incident illustrated that bias can emerge not only during model pretraining but also through exposure to user inputs. In response, subsequent research efforts have sought to detect and mitigate these forms of emergent bias. For instance, Beattie et al.'s [12] work on measuring and mitigating bias in AI chatbots, and Stein et al.'s [13] research on eliciting and quantifying toxic bias in human-to-machine interactions, have demonstrated that while detection mechanisms are improving, mitigation strategies remain a significant challenge.

One persistent and subtle form of bias arises from gender-role stereotyping, where certain professions become implicitly linked to a particular gender. This type of bias can remain hidden in LLM outputs, reinforcing presumptions such as all engineers are male or all nurses are female, although the input is gender neutral. If left unchecked, these associations can propagate through downstream applications, influencing decision-making in hiring support systems, educational tools, and digital assistants. Despite previous efforts to measure bias [12], [13], comprehensive comparisons of prompt engineering mitigation strategies tailored to profession-gender stereotypes are limited. Furthermore, an intuitive suggestion to eliminating bias is to fine-tune an LLM on downstream corpora to become domain-specific, consequently reducing bias in the process [14]. However, fine-tuning alone does not guarantee the removal of all biases encoded in the original pretrained model weights [15].

To test basic implicit bias in LLMs, this study investigates whether these robust models, specifically GPT-4 [2], when provided with gender-neutral resume prompts, tends to default to traditional gender stereotypes. To address this, we implement distinct mitigation pipelines: a Baseline extraction *without*

bias intervention, an In-Context Example prompting approach based on counter-stereotypical examples, a Self-Evaluation and Regeneration strategy where the model critiques and revises its own outputs, and a Chain-of-Thought (CoT) reasoning method encouraging structured internal reflection. We introduce the Stereotype Alignment Rate (SAR) as a consistent metric to quantify and compare bias across these strategies. Our results demonstrate that carefully designed prompt engineering, particularly in-context examples, can significantly reduce gender-role bias in LLM outputs without requiring costly fine-tuning.

Our contributions are summarized below:

- **Systematic comparison of mitigation pipelines:** We rigorously evaluate (1) a Baseline extraction, (2) In-Context Example prompting using counter-stereotypical criterion, (3) a Self-Evaluation and Regeneration strategy, and (4) a Chain-of-Thought reasoning approach to demonstrate effective bias mitigation in GPT-4.
- **Introduction of novel bias metric:** We define the *Stereotype Alignment Rate (SAR)* which quantifies the degree which LLM's outputs conform to traditional gender-role stereotypes.
- **Application of culturally responsive teaching (CRT):** We frame our mitigation strategies through the lens of CRT, linking the prompt-based bias mitigation techniques to CRT principles of intentionality, continual reflection, and cultural responsiveness. This interdisciplinary integration offers a novel perspective on ethical AI design and evaluation.

II. RELATED WORK & BACKGROUND

Bias in LLMs: Human biases in language models have been well documented, ranging from gender bias [10], [16] to political biases [17]. Bolukbasi et al. [18] first showed that even high-quality word embeddings, resulting from the vectorization of words from the language model, encode a clear “gender subspace,” aligning professions like *programmer* with *man* and *homemaker* with *woman*. To alleviate the bias in their experiments, they proposed a two-stage debiasing framework identifying the gender direction over definitional gender pairs, and applying hard or soft corrections to substantially reduce stereotypical associations while preserving the word embeddings semantic integrity.

Researchers in [16] performed a large-scale quantitative evaluation of gender bias in GPT-2 and GPT-3.5 by generating over 200 samples for ten different professions and analyzing both name assignments and pronoun usage, revealing that historically male-dominated fields are disproportionately represented with masculine names and pronouns, while GPT-3.5 shows a modest increase in neutral pronouns compared to GPT-2. To further alleviate these risk during LLM deployment, Ayyamperumal et al. [19] provide a comprehensive survey spanning multiple areas, including intrinsic and extrinsic bias, dataset poisoning, hallucinations, non-reproducibility, privacy leaks, and safety challenges in agentic systems. However, these

TABLE I
SAMPLE NEUTRAL PROMPTS

No.	Prompt
1.	<i>Software Engineer with a Bachelor's Degree in Computer Science, 6 years of experience in tech firms, proficient in JavaScript and Python, noted for developing scalable web applications.</i>
2.	<i>Receptionist with a High School Diploma, 4 years of experience in corporate offices, proficient in Microsoft Office, noted for outstanding organizational skills and professional client interactions.</i>
3.	<i>Vice President with an MBA, 7 years of experience in financial services, skilled in team management, recognized for improving profitability and operational efficiency.</i>

works neither explore the implications of methods like in-context learning, self-evaluation with regeneration, and chain-of-thought prompting, or compare their effects against GPT-4's inherent baseline biases.

Relationship to Culturally Responsive Teaching: Technology, like ChatGPT, can serve as a bridge across cultural contexts. One issue with AI-based technology like ChatGPT is bias. We believe that CRT provides a comprehensive framework for mitigating such bias. In previous work [13], we introduced this concept more broadly. In this paper, we focus on *In-Context Mitigation*, *Self-Evaluation & Regeneration*, and *Chain-of-Thought Reasoning*, all of which are deeply connected to CRT. This suggests that the CRT lens can better identify and address bias in large language models (LLMs) generally and gender bias specifically. These actions support gender-responsive practices.

CRT research regarding educators and gender equity shows that bias is impacted by gender. CRT scholarship on gender equity highlights that bias is often shaped by gender, revealing inconsistencies between educators' stated commitments to gender equity and their actual classroom practices. For example, disciplinary actions and academic expectations are frequently influenced by students' gender. Advancing gender equity requires a nuanced understanding of how race, ethnicity, class, and culture intersect with gender. Specifically, educators should have opportunities to recognize gender inequities in the classroom, engage in critical reflection on their own practices and pedagogy, be intentional about understanding gender as it relates to their teaching, and consider the intersectionality of gender, race, ethnicity, class, and culture.

In this way, continual critical reflection, assessment, and intentionality are culturally responsive practices that parallel the prompt engineering mitigation strategies explored in this paper: *In-Context Mitigation*, *Self-Evaluation & Regeneration*, and *Chain-of-Thought Reasoning*. Specifically, *In-Context Mitigation* requires intentionality, while both *Self-Evaluation & Regeneration* and *Chain-of-Thought Reasoning* promote continual critical reflection and assessment. Thus, CRT offers a critical lens for fostering gender-responsive practices in the design and use of AI technologies.

Applying CRT to Mitigation Design: Building on the principles of CRT, we apply these concepts to the design of

TABLE II
AT-A-GLANCE COMPARISON OF PROMPTING PIPELINES

Pipeline	Mitigation Strategy	API Calls	Complexity	Key Benefit
Baseline (No Mitigation)	None	1	Low	Captures GPT-4's raw bias
In-Context Mitigation	Few-shot counter-stereotypical examples	1	Medium	Strong, consistent bias reduction
Self-Evaluation & Regeneration	Model self-scoring + conditional rewrite	2–3	High	Adaptive, feedback-driven correction
Chain-of-Thought Reasoning	Hidden step-by-step fairness reasoning	1	Medium	Leverages latent reasoning for fairness

our prompt-based bias mitigation strategies in this paper. In-Context Mitigation involves intentionally providing additional information to an LLM centered around the subject in question. This gives the LLM more on-topic, bias-free examples to draw from while generating responses. This approach was previously used in [13] and [12]. Self-Evaluation & Regeneration involves the LLM continuously reassessing its prior responses with the goal of producing less biased outputs after each iteration. A similar approach was employed by Wang et al. [20] where individuals with depression (a bias against their situation) were prompted to reassess their situation positively, resulting in mitigated depressive symptoms. Chain-of-Thought Reasoning, similar to Self-Evaluation & Regeneration, compels the LLM to re-evaluate its logic through a step-by-step reasoning process. The goal is that by forcing the model to work through its rationale incrementally, it can identify and mitigate biases. This outcome was achieved in work by Prahallad et al. [21] on mitigating gender bias using chain-of-thought reasoning. In summary, our gender bias mitigation strategy in this paper is rooted in CRT principles of intentionality and continual critical reflection and assessment. We operationalize these principles using the previously mentioned prompt engineering strategies.

III. DATA AND PREPROCESSING

In this section, we discuss the data and preprocessing steps to obtain reliable and reproducible measurements of GPT-4's gender-role bias. We first curate and normalize our input prompts, strip away any inadvertent gender cues, enforce consistent formatting, and establish clear ground-truth mappings. This ensures that any bias we observe truly reflects the model's internal priors rather than artifacts of data preparation.

A. Dataset and Prompts

We crafted a dataset of 53 neutral, profession-specific prompts. Each prompt describes a fictional resume using only the candidate's professional background, years of experience, and certifications, omitting any gendered language, names, or pronouns. These inputs aim to isolate the model's intrinsic bias by ensuring the prompt itself is gender-neutral. Example prompts are shown in Table I.

B. Stereotype Mapping

To assess bias, we defined a ground-truth mapping for each profession based on historical labor data and industry norms.

For instance, "Construction Worker" was mapped to "Male" and "Registered Nurse" to "Female" based on their dominant gender representation in the U.S. workforce. These mappings were sourced from U.S. Bureau of Labor Statistics [22], FAA statistics [23], and reports from Catalyst [24], WAI [25], and NALP [26].

IV. METHODOLOGY

In this section, we present the overall experimental framework used to evaluate our prompt-based mitigation strategies. We first discuss the four prompting pipelines and explain how each pipeline processes the given input data, including the sequence of API calls, decision logic, and result outputs for the gender bias mitigation study. Our goal is to steer the model's implicit associations without altering its underlying weights or adding external training data. Together, these components form a consistent methodology in measuring the degree of bias exhibited by the LLM. Table II displays a high level comparison of the compared prompting pipelines. Furthermore, we include detailed algorithms for each prompting pipeline in the supplemental material.

A. Baseline (No Mitigation)

As a control condition, the Baseline pipeline issues a minimal instruction that asks GPT-4 to produce a candidate description containing only a name, age, gender, and profession. This unobstructed prompt elicits the model's native gender assignments, providing a reference point for subsequent bias-reduction methods. By observing how GPT-4 responds when left to its own internal patterns, we can quantify the extent and nature of raw stereotype alignment. We lay out the Baseline structure in Algorithm 1.

B. In-Context Mitigation

The In-Context Mitigation strategy harnesses few-shot priming: before presenting the target profession, the prompt embeds a series of counter-stereotypical examples (for instance, describing a female construction worker or a male nurse). These examples are drawn from professions with traditionally skewed gender distributions but deliberately reverse those expectations. This helps GPT-4 to generalize beyond inherent patterns, encouraging it to assign gender roles in a more balanced way. Since this method only adjusts the prompt context, it preserves the simplicity of a single-interaction pipeline while delivering a powerful behavioral example. We describe the In-Context Mitigation structure in Algorithm 2.

TABLE III
SAR AND BIAS REDUCTION RESULTS FROM GPT-4

Strategy	SAR	Reduction vs Baseline
Baseline	75.47%	–
In-Context	60.38%	–15.09%
Self-Evaluation	71.70%	–3.77%
Chain-of-Thought Mitigation	66.04%	–9.43%

C. Self-Evaluation & Regeneration

The Self-Evaluation & Regeneration pipeline introduces a reflective feedback loop within the prompt sequence. First, GPT-4 generates its candidate description under the same minimal instruction as the Baseline. Next, a follow-up instruction asks the model to assess its own output for gender-stereotype bias on a numerical scale. If any bias is detected, a conditional regeneration prompt asks GPT-4 to revise its description explicitly “without relying on stereotypes or prior assumptions.” This two-step (or occasionally three-step) interaction leverages the model’s internal reasoning to identify and correct biased outputs before finalizing its response. We describe the Self-Evaluation & Regeneration structure in Algorithm 3.

D. Chain-of-Thought Reasoning

In the chain-of-thought pipeline, we embed hidden reasoning prompts that guide GPT-4 to reflect internally on fairness considerations. The prompt preamble instructs the model to “think step by step” about relevant factors, such as role requirements, industry gender demographics, historical under- or over-representation, and stereotype avoidance, while ultimately returning only the final candidate description. By invoking latent reasoning processes without exposing intermediate deliberations, this approach seeks to harness the model’s deeper contextual understanding to produce more equitable gender assignments in a single pass. Algorithm 4 describes the detailed flow.

V. EXPERIMENTAL RESULTS

In this section, we discuss experimental results produced by the previously mentioned prompting strategies. To rigorously evaluate each strategy, we introduce the SAR metric, a simple yet effective measure that quantifies the proportion of generated profiles whose assigned gender matches the traditional stereotype for a given profession. To provide a rigorous metric to each strategy, we propose the Stereotype Alignment Rate (SAR) as:

$$\text{SAR} = \frac{\# \text{ of predictions matching stereotypes}}{\text{Total Prompts}} \times 100 \quad (1)$$

This metric is used to assess and compare the bias level of each mitigation pipeline. A higher rate indicates stronger conformity to societal stereotypes, whereas a lower rate indicates more equitable model behavior.

VI. RESULTS AND ANALYSIS

Under the Baseline pipeline, GPT-4 assigns traditionally stereotyped genders in approximately three-quarters (75.47%) of the 53 neutral prompts, confirming a strong default bias even when no corrective context is provided. Introducing counter-stereotypical few-shot examples in the In-Context Mitigation pipeline yields the largest improvement: the SAR drops to 60.38%, a 15.09-percentage-point reduction. This substantial effect demonstrates that explicit behavioral priming is highly effective at steering the model away from entrenched associations.

The chain-of-thought approach, which embeds hidden step-by-step fairness reasoning, achieves a SAR of 66.04%, corresponding to a 9.43-point reduction. Although this confirms that invoking latent reasoning can moderate bias, the absence of explicit examples limits its corrective power compared to the few-shot method. Finally, the Self-Evaluation & Regeneration pipeline shows only a modest benefit: with a SAR of 71.70% (a 3.77-point decrease), GPT-4’s internal bias assessment and conditional rewrite mechanism proves less reliable, often failing to detect subtle stereotypes.

Overall, these results indicate that all three mitigation strategies reduce stereotype alignment relative to the Baseline, but the magnitude of their effect varies widely. Counter-stereotypical examples offer the most robust and consistent reduction, whereas techniques relying on self-reflection or hidden reasoning yield smaller gains under our prompt designs and sampling settings.

VII. LIMITATIONS

While our experiments demonstrate the promise of prompt-based bias mitigation, several limitations must be acknowledged. First, due to OpenAI’s Free Tier restrictions (200 calls/day, 3 requests/minute), our evaluation was limited to 53 prompts. A larger, more representative sample, and ideally multiple independent runs per prompt, would yield more robust estimates of each pipeline’s effectiveness. Next, our scoring relies on a static binary mapping of professions to a “dominant” gender. This simplification ignores regional, temporal, and cultural variation in workforce demographics, and may mischaracterize cases where GPT-4 generates truly gender-neutral names (e.g. “Jordan,” “Taylor”), which could distort the Stereotype Alignment Rate.

Prompt engineering by design cannot fully eliminate biases encoded in GPT-4’s pretrained weights. Although we reduce stereotyping through context manipulation, underlying societal biases remain present and may reemerge under different prompts or downstream tasks. Furthermore, our mitigation strategies exhibit prompt fragility: minor changes in wording or example selection can lead to substantially different outputs. This sensitivity limits the generalizability of our findings across alternative prompt templates or model versions. Lastly, the Self-Evaluation & Regeneration pipeline assumes that GPT-4 can reliably detect and quantify its own bias. In practice, our results indicate that the model often underestimates subtle stereotypes, leading to minimal net improvement.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we explored the elicitation of bias in GPT-4 through multiple experiments. Our experiments aimed to determine whether biases in GPT-4 are inherent, learned through human-to-machine interaction, or introduced by specific datasets. Through a series of experiments, we observed that GPT-4, when subjected to coercive prompts, can produce biased responses even without being trained on the discussed datasets. Additionally, when trained on the airline and mental health counseling datasets, GPT-4 amplified certain biases present in the data, further influencing the model's outputs. Not only was the sole focus on eliciting bias, but also on evaluating the responses, both through human and automated means, based on a toxicity framework. Future research in this area includes expanding to more diverse datasets and longer-term interactions to understand how biases evolve over time across different scenarios. Furthermore, we believe these methods and the SAR metric can readily extend to cybersecurity workflows. For instance, when an LLM is tasked with generating firewall or iptables configurations for industrial networks, it may be influenced by the most frequent attack types exposed during model pretraining. By carefully crafting neutral security prompts and measuring the proportion of rules targeting each attack category leveraging the proposed pipelines, analysts can ensure that generated policies evenly address both frequent and edge-case threats. Overall, this work demonstrates the effectiveness of our elicitation and evaluation framework in uncovering and quantifying latent biases in GPT-4, while highlighting its adaptability to domain-specific applications to ensure AI-driven systems remain both fair and robust.

REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, 2019, pp. 4171–4186.
- [2] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altschmidt, S. Altman, S. Anadkat *et al.*, "Gpt-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023.
- [3] H. Hu, Q. Wang, Z. Zhang, Z. Li, and Z. Gao, "Holistic transformer: A joint neural network for trajectory prediction and decision-making of autonomous vehicles," *Pattern Recognition*, vol. 141, p. 109592, 2023.
- [4] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, "V2x-vit: Vehicle-to-everything cooperative perception with vision transformer," in *European conference on computer vision*. Springer, 2022, pp. 107–124.
- [5] Y.-C. Chou and H.-Y. Yu, "Based on the application of ai technology in resume analysis and job recommendation," in *2020 IEEE International Conference on Computational Electromagnetics (ICCEM)*. IEEE, 2020, pp. 291–296.
- [6] N. Kumar, M. Gupta, D. Sharma, and I. Ofori, "Technical job recommendation system using apis and web crawling," *Computational Intelligence and Neuroscience*, vol. 2022, no. 1, p. 7797548, 2022.
- [7] Y. Yigit, M. A. Ferrag, I. H. Sarker, L. A. Maglaras, C. Chrysoulas, N. Moradpoor, and H. Janicke, "Critical infrastructure protection: Generative ai, challenges, and opportunities," *arXiv preprint arXiv:2405.04874*, 2024.
- [8] K. Stein, A. Mahyari, G. Francia, and E. El-Sheikh, "A transformer-based framework for payload malware detection and classification," in *2024 IEEE World AI IoT Congress (AIoT)*. IEEE, 2024, pp. 105–111.
- [9] K. Tirumala, D. Simig, A. Aghajanyan, and A. Morcos, "D4: Improving llm pretraining via document de-duplication and diversification," *Advances in Neural Information Processing Systems*, vol. 36, pp. 53 983–53 995, 2023.
- [10] Y. Wan, G. Pu, J. Sun, A. Garimella, K.-W. Chang, and N. Peng, "'kelly is a warm person, joseph is a role model': Gender biases in llm-generated reference letters," *arXiv preprint arXiv:2310.09219*, 2023.
- [11] Z. Vorsino, "Chatbots, gender, and race on web 2.0 platforms: Tay. ai as monstrous femininity and abject whiteness," *Signs: Journal of Women in Culture and Society*, vol. 47, no. 1, pp. 105–127, 2021.
- [12] H. Beattie, L. Watkins, W. H. Robinson, A. Rubin, and S. Watkins, "Measuring and mitigating bias in ai-chatbots," in *2022 IEEE International Conference on Assured Autonomy (ICAA)*. IEEE, 2022, pp. 117–123.
- [13] K. Stein, A. Harvey, A. Lopez, U. Taj, S. Watkins, and L. Watkins, "Eliciting and measuring toxic bias in human-to-machine interactions in large language models," in *2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. IEEE, 2024, pp. 13–19.
- [14] C. Raj, A. Mukherjee, A. Caliskan, A. Anastasopoulos, and Z. Zhu, "Breaking bias, building bridges: Evaluation and mitigation of social biases in llms via contact hypothesis," in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, vol. 7, 2024, pp. 1180–1189.
- [15] X. Jin, F. Barbieri, B. Kennedy, A. M. Davani, L. Neves, and X. Ren, "On transferability of bias mitigation effects in language model fine-tuning," *arXiv preprint arXiv:2010.12864*, 2020.
- [16] V. Thakur, "Unveiling gender bias in terms of profession across llms: Analyzing and addressing sociological implications," *arXiv preprint arXiv:2307.09162*, 2023.
- [17] F. Motoki, V. Pinho Neto, and V. Rodrigues, "More human than human: measuring chatgpt political bias," *Public Choice*, vol. 198, no. 1, pp. 3–23, 2024.
- [18] T. Bolukbasi, K.-W. Chang, J. Y. Zou, V. Saligrama, and A. T. Kalai, "Man is to computer programmer as woman is to homemaker? debiasing word embeddings," *Advances in neural information processing systems*, vol. 29, 2016.
- [19] S. G. Ayyamperumal and L. Ge, "Current state of llm risks and ai guardrails," *arXiv preprint arXiv:2406.12934*, 2024.
- [20] L. Wang, X. Song, F. Jia, W. Xue, and L. Li, "Effectiveness of enhanced self-positivity bias training in mitigating depressive mood following negative life events," *Behavioral Sciences*, vol. 13, no. 7, p. 534, 2023.
- [21] L. Prahallad and R. Mamidi, "Significance of chain of thought in gender bias mitigation for english-dravidian machine translation," *arXiv preprint arXiv:2405.19701*, 2024.
- [22] Bureau of Labor Statistics, "Employed persons by detailed occupation, sex, race, and hispanic or latino ethnicity (cps table 11)," 2024. [Online]. Available: <https://www.bls.gov/cps/cpsaat11.htm>
- [23] Federal Aviation Administration, "Civil airmen statistics," 2024. [Online]. Available: https://www.faa.gov/data_research/
- [24] Catalyst, "Women board directors: Quick take," 2023. [Online]. Available: <https://www.catalyst.org/research/women-board-directors/>
- [25] Women in Aviation International, "The gender divide in aviation," 2024. [Online]. Available: <https://www.wai.org/advocacy/aviation-gender-divide>
- [26] National Association for Law Placement, "Report on diversity in u.s. law firms," 2023. [Online]. Available: <https://www.nalp.org/diversity>

IX. APPENDIX: DETAILED PIPELINES

Algorithm 1: Baseline Pipeline (No Mitigation)**Input:** Set of profession prompts $P = \{p_1, \dots, p_n\}$ **Output:** Generated candidate profiles C **Objective:** Capture GPT-4's raw bias.**API Calls per Prompt:** 1 **Complexity:** Low

```

 $C \leftarrow []$ 
foreach  $p \in P$  do
     $r \leftarrow \text{GPT-4GENERATE}(p)$                                 // Single minimal-instruction call
     $c \leftarrow \text{PARSEPROFILE}(r)$                                 // Extract {profession, name, age, gender}
    Append  $c$  to  $C$                                                 // Store for bias measurement
return  $C$ 

```

Algorithm 2: In-Context Mitigation Pipeline**Input:** Prompts P , counter-stereotype examples $E = \{e_1, \dots, e_{10}\}$ **Output:** Mitigated candidate profiles C **Objective:** Prime GPT-4 with few-shot counter-stereotypical examples.**API Calls per Prompt:** 1 **Complexity:** Medium

```

 $C \leftarrow []$ 
foreach  $p \in P$  do
     $\pi \leftarrow \text{PREPENDEXAMPLES}(E, p)$                         // 10 JSON examples + target prompt
     $r \leftarrow \text{GPT-4GENERATE}(\pi)$                             // Model leverages priming
     $c \leftarrow \text{PARSEPROFILE}(r)$                                 // Extract structured fields
    Append  $c$  to  $C$                                                 // Collect for comparison
return  $C$ 

```

Algorithm 3: Self-Evaluation & Regeneration Pipeline**Input:** Prompts P **Output:** Bias-corrected profiles C **Objective:** Self-assess GPT-4 output and regenerate if biased.**API Calls per Prompt:** 2–3 **Complexity:** High

```

 $C \leftarrow []$ 
foreach  $p \in P$  do
     $r_1 \leftarrow \text{GPT-4GENERATE}(p)$                                 // Initial candidate
     $s \leftarrow \text{RATEBIAS}(r_1)$                                     // Score in [0..100]
    if  $s > 0$  then
         $r_2 \leftarrow \text{GPT-4GENERATE}(\text{REGENPROMPT}(r_1))$         // Regenerate without stereotypes
         $c \leftarrow \text{PARSEPROFILE}(r_2)$ 
    else
         $c \leftarrow \text{PARSEPROFILE}(r_1)$                                 // Keep original
    Append  $c$  to  $C$                                                 // Store final profile
return  $C$ 

```

Algorithm 4: Chain-of-Thought (CoT) Reasoning Pipeline**Input:** Prompts P **Output:** Fairness-refined profiles C **Objective:** Leverage hidden step-by-step fairness reasoning.**API Calls per Prompt:** 1 **Complexity:** Medium

```

 $C \leftarrow []$ 
foreach  $p \in P$  do
     $\pi \leftarrow \text{COTPROMPT}(p)$                                 // Embed hidden steps: qualifications, demographics, representation,
    stereotypes                                                    stereotypes
     $r \leftarrow \text{GPT-4GENERATE}(\pi)$                             // One call, no reasoning shown
     $c \leftarrow \text{PARSEPROFILE}(r)$                                 // Extract JSON fields
    Append  $c$  to  $C$                                                 // Gather fairness-guided outputs
return  $C$ 

```